Predict New York City Airbnb Prices Using Linear Regression

Project overview

This project explores the AB_NYC_2019 Airbnb listings dataset and builds a Linear Regression model to predict listing price. It demonstrates data auditing, cleaning, outlier treatment, categorical encoding, model training, and evaluation.

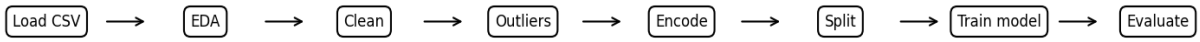
Objective

Predict listing prices from available listing features and assess the model with R^2, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

Dataset

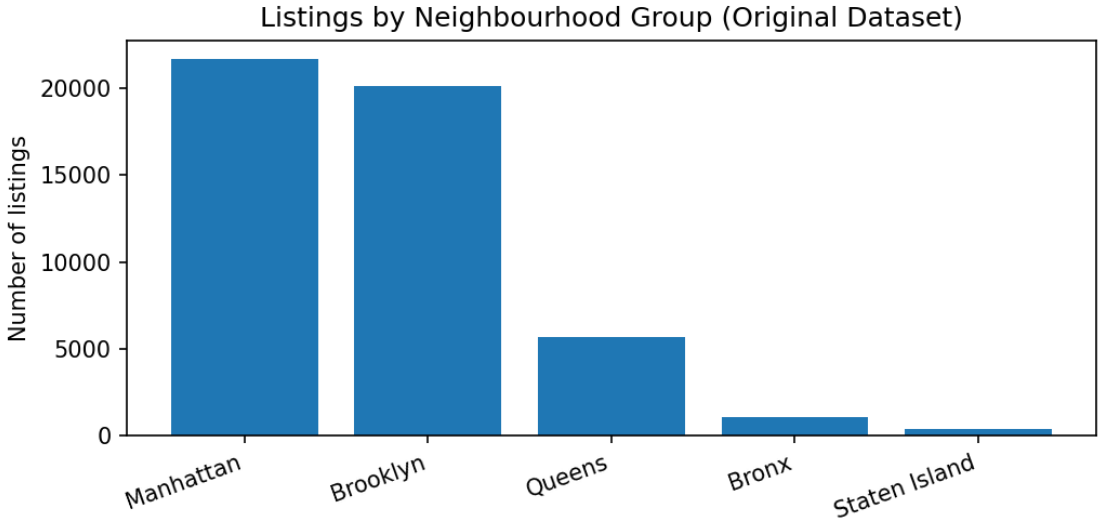
The commonly used AB_NYC_2019 dataset contains 48,895 rows and 16 columns before cleaning. Fields include neighbourhood group, room type, price, minimum nights, review counts, and availability. Verify your downloaded file's exact shape by running the notebook.

Project workflow

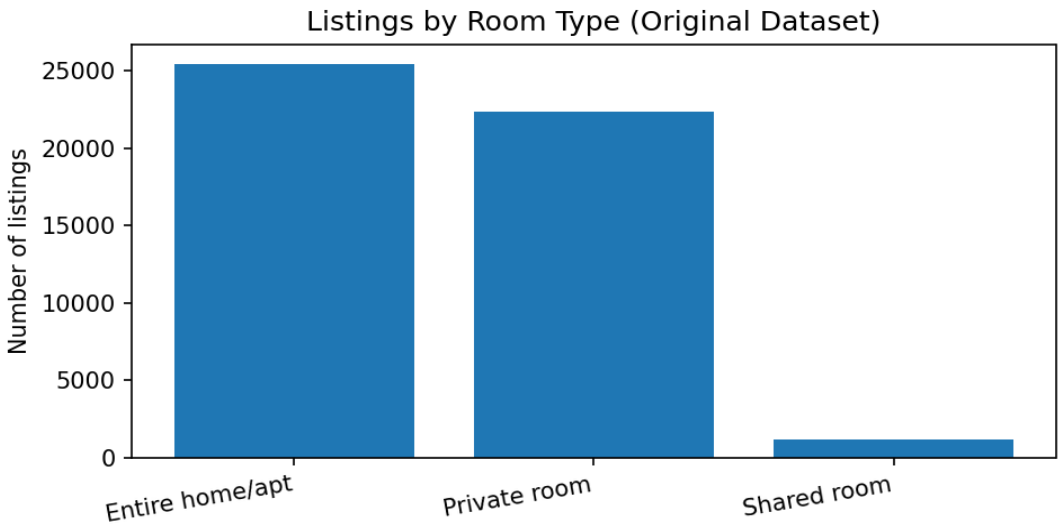


Exploratory Data Analysis (EDA)

The notebook checks shape, column names, data types, missing-value counts and percentages, and numeric summaries. These bar graphs show the original dataset's listing distribution before cleaning.



Methodology, Graphs, and Conclusion



Data cleaning

Rows with missing or non-positive prices are removed. Missing reviews_per_month values are set to zero; last_review is filled with a placeholder and then excluded from model features. Identifier, name, host, coordinates, and high-cardinality neighbourhood fields are removed. Remaining numeric missing values are filled with medians and categorical missing values with the mode.

Outlier handling and encoding

Prices above the 95th percentile and listings with minimum_nights of 365 or more are excluded. neighbourhood_group and room_type are one-hot encoded with drop_first=True. The prediction target is price.

Training and evaluation

The dataset is split into 80% training and 20% testing data using random_state=42. Linear Regression is trained on the training set. The code prints R^2, MAE, and RMSE, then plots actual versus predicted prices and residuals. It also compares MAE for lower-priced and higher-priced listings.

Results — fill in after running the notebook

R^2: _____ MAE: _____ RMSE: _____

Lower-priced MAE: _____ Higher-priced MAE: _____

Graph interpretation

Points close to the 45-degree line in the actual-versus-predicted plot indicate more accurate predictions. In the residual plot, a random spread around zero is desirable. Different MAE values between price groups indicate different levels of prediction error.

Conclusion

This project presents an end-to-end regression workflow for Airbnb price prediction. Report model performance using the values produced by your own run. Linear Regression is a useful baseline, but may not capture nonlinear price patterns or all location and property differences. Future work could compare tree-based models, add appropriate location features, and use cross-validation.